A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

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Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in Computer Systems & Network Engineering

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate dissertation under my supervision.

Signature of the Supervisor: ..... Date 24/8/25

Signature of the Co-Supervisor: ..... Date 24/08/2025

Acknowledgement

Working on this final-year research project has been an immensely enriching academic experience for me. Through the process of developing my research proposal, I have gained valuable insights into the research process, including how to formulate research questions, design suitable methodologies, and anticipate potential challenges.

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I look forward to continuing to develop my research skills and contribute meaningful insights through this project.

Abstract

Low-power and lossy networks (LLNs) supporting the Internet of Things (IoT) often utilize the IPv6 routing protocol for low-power and lossy networks (RPL). While RPL provides scalability and energy efficiency, its lightweight architecture makes it vulnerable to internal attacks such as sinkhole and selective forwarding. Existing defenses are either fragmented, targeted at specific attacks, or require extensive computing power, making them unsuitable for resource-constrained IoT devices.

To overcome these limitations, our study proposes a lightweight integrated trust and security framework for IoT networks based on RPL. The framework comprises four interacting modules: a Trust Computation Engine that evaluates node trustworthiness based on residual energy, temperature, historical behavior, and trust accuracy; an Intrusion Detection System that inspects forwarding ratios, parent-child relationship changes, control message anomalies, and routing inconsistencies to simultaneously detect sinkhole and selective forwarding attacks in real time, providing anomaly alerts with node ID and attack type for secure routing; a modified RPL objective function that incorporates trust and security metrics for adaptive parent selection; and a Resilience Module that provides node isolation, mitigation, and rapid recovery. The framework will be implemented and tested on Contiki-NG OS using the Cooja simulator.

The impact of the proposed framework will be assessed based on detection accuracy, false positive rate, packet delivery rate, energy and memory consumption, and recovery time. By combining trust assessment and double-attack detection within a lightweight architecture, the framework aims to improve the resilience of IoT networks while maintaining usability for devices with limited capabilities. Potential applications include disaster response, smart city infrastructure, and the Industrial Internet of Things, providing a scalable and commercially viable solution.

Keywords: IoT Security, RPL Security, Trust Management, Intrusion Detection, Lightweight Framework

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1.0 Introduction

The Internet of Things (IoT) has become a cornerstone of modern digital infrastructure, supporting areas such as healthcare, disaster management, industrial automation, and smart cities. These applications rely on networks of resource-constrained devices that must operate reliably under severe constraints on processing power, memory, and power consumption. Therefore, ensuring secure and reliable communications in such low-power and loss-prone networks is a critical research challenge [1].

To address these limitations, the IPv6 Routing Protocol for Low-Power and Lossy Networks (RPL) has been standardized as the de facto routing protocol for IoT environments. RPL enables scalable data forwarding and efficient device orchestration; however, its lightweight architecture also creates vulnerabilities to routing-layer attacks such as sinkhole and selective forwarding. These attacks can seriously degrade packet delivery, compromise network reliability, and threaten the trust in IoT systems deployed in mission-critical environments [17].

This study aims to address these vulnerabilities by integrating trust management and intrusion detection into a unified framework for RPL-based IoT networks. Instead of using fragmented approaches, the proposed framework combines simplified trust assessment, real-time routing attack detection, adaptive routing decisions, and resilience mechanisms into a holistic design. By enhancing routing resilience to evolving threats, the framework aims to improve the reliability of IoT deployments in mission-critical scenarios such as disaster response, healthcare monitoring, and smart city infrastructure [22].

1.1 Background & Literature Review

1.1.1 Background

The Internet of Things (IoT) connects many resource-constrained devices sensors, actuators, and embedded systems over low-power lossy networks (LLNs) characterized by unstable links, limited bandwidth, and a strict power budget [1], [22]. To support such environments, the IPv6 routing protocol for low-power lossy networks (RPL) has been standardized as the de facto routing solution for resource-constrained IoT deployments. RPL structures node into a destination-oriented directed acyclic graph (DODAG) rooted at an edge router and computes ranks using objective functions (e.g., OF0, MRHOF) to guide parent node selection and upstream/downstream routing. RPL uses control messages DIO (DODAG Information Object), DIS (DODAG Information Request), and DAO (Destination Advertisement Object) to form and maintain the topology [17], [22]. While this architecture provides scalability and energy awareness, it also opens the door to attacks at the routing

level, as decisions depend on locally observable metrics and trust in neighbor reports in limited, intermittently connected networks [1], [17], [22].

Routing level threats exploit these characteristics. In a sinkhole attack, a compromised node forges or publishes deceptively attractive routing information to attract traffic from neighbors, then drops, delays, or modifies packets, degrading delivery and distorting parent node selection [4-6]. In selective forwarding, an attacker participates in routing but discards a subset of data packets (e.g., by flow, destination, or rate), creating subtle but persistent losses that are difficult to diagnose [10], [15], [20]. These attacks reduce the packet delivery ratio (PDR), increase latency, and destabilize the DODAG structure by causing frequent parent node changes and reranking events [18]. Additional manipulations, such as rank irregularity, can further distort path costs and exacerbate the disruption when combined with packet failures or selective packet drops [11], [17]. Because insider attackers can appear compliant at the packet format level, pure cryptographic compliance does not prevent such misbehavior at the behavioral level [17], [22].

Recent studies also confirm that sinkholes and selective forwarding are the two most damaging and prevalent routing attacks in RPL networks. Prajapati et al. [1] reported that together they account for more than 60–70% of documented routing layer degradations in IoT testbeds. Raof et al. [17] showed that sinkhole attacks can reduce the packet delivery ratio (PDR) by more than 50%, while selective forwarding causes 20–40% persistent packet loss, depending on the intensity. Alfrichat et al. [19] further noted that ≈65% of RPL security research focuses on these two attacks, reflecting their dominance in both experimental and simulation studies. Verma and Ranga [22] experimentally demonstrated that the combined effect of sinkhole and selective forwarding can reduce the PDR from 98% to only 30–40%, destabilizing DODAG more than other threats such as Sybil or rank manipulation. These results justify focusing this study on these two attacks, ensuring both coverage of the most serious threats and consistency with established research practices.

The operational limitations of IoT nodes exacerbate the problem. Devices typically have limited processors, memory, and battery resources, limiting the ability to implement resource-intensive security features and continuous monitoring [1], [19], [22]. Control plane activity should be minimal to avoid overloading the already constrained wireless environment and conserve energy for application traffic [22]. Many effective security techniques in general-purpose IP networks (e.g., complex cryptographic handshakes, deep learning models with large feature sets, or centralized analytics with high reporting overhead) become impractical in LLNs due to high processing, memory, and power costs, as well as unstable connectivity [1], [19], [22]. Moreover, timely detection is critical: delayed response allows attackers to corrupt parent node selection and exacerbate congestion or partitioning, after which recovery requires additional energy and time [17], [18], [22].

In this context, three families of security tools define the problem space at a high level. (i) Cryptographic and access control mechanisms authenticate identities and messages, reducing external interference, but cannot guarantee honest forwarding, leaving malicious behavior of internal users (e.g., selective dropping) unchecked [17], [22]. (ii) Trust-based strategies maintain the behavioral reputation of neighbors, determined by observations such as forwarding success, stability, or energy trends, and use it to influence the selection of parent nodes. Despite their conceptual simplicity, they must avoid instability (e.g., oscillations caused by rapid trust fluctuations) and false positives in dynamic topologies [1], [19], [23]. (iii) Intrusion detection approaches monitor traffic and control plane behavior to detect anomalies indicative of attacks (e.g., abnormal forwarding ratios, packet synchronization violations, sequence or routing mismatches), but they must balance detection sensitivity with computational and communication costs to remain feasible on nodes with limited capabilities [19], [22].

The LLN measurement context introduces additional challenges. In lossy multicast topologies, benign losses, cycling, and bursty interference can mimic attack symptoms, making it difficult to distinguish between normal dynamics and malicious behavior [18], [22]. Metrics such as control to data forwarding ratios, inter packet timing patterns, and sequence continuity can be informative, but require careful thresholding and aggregation over time to smooth out transient effects and avoid oversensitivity to short-term connection variability [18], [19], [22]. Similarly, routing consistency indicators (e.g., consistency between advertised path costs and observed forwarding results) can detect deception attempts, but capturing them in a lightweight form without heavy global state remains challenging in distributed LLNs [17], [22]. Therefore, the tradeoff is fundamental: stronger detection typically increases processing, memory, and messaging overhead, which reduces battery life and channel availability; weaker detection increases the risk of missed attacks or slow responses [1], [19], [22].

The realities of evaluation also pose a challenge. IoT security research often relies on simulations (e.g., Contiki-NG/Cooja) to study the scale and conditions of attackers, measuring PDR, latency, power/CPU consumption, false positives/negatives, and recovery time in controlled scenarios [18], [22]. While machine learning-based detectors can achieve high accuracy in some settings, they often assume feature richness or label availability that are difficult to guarantee in RPL settings; training and distributing models further increases the computational and communication burden, which is incompatible with constrained nodes [3], [19], [21], [22]. Consequently, there is a continuing need for low overhead, behaviorally informed indicators and lightweight aggregation schemes that enable timely detection without exceeding the resource budgets typical for RPL deployments [19], [22].

From a systems perspective, maintaining reliable routing in adversarial environments requires consistency between (a) how misbehavior is observed, (b) how trust or anomaly scores are formed, and (c) how routing decisions respond. If detection results are not reflected in parent node selection, attacks can continue to impact routes; if routing overreacts to noisy indicators, networks suffer from instability and energy loss due to churn [17], [22], [23]. Similarly, in addition to detection, mitigation and recovery measures are required, including temporary isolation of suspicious nodes and subsequent reintegration after behavioral normalization, to preserve connectivity and prevent permanent fragmentation while minimizing the cost of false isolation [17], [22], [23]. Achieving this balance under tight energy and computational constraints is critical for practical RPL based IoT security [1], [19], [22].

In summary, the lightweight RPL architecture and the realities of LLNs create a behavioral attack surface where failures and selective forwarding can silently undermine delivery and stability while maintaining control plane compliance [4-6], [10], [15], [17], [20], [22]. Therefore, defenses must: (1) operate within strict CPU/memory/energy constraints, (2) rely on low-overhead indicators that are robust to favorable dynamics, (3) support timely node-level detection that informs routing choices, and (4) include mitigation and recovery to maintain service under attack [1], [17-19], [22], [23]. This background frames the existing problem addressed in this study: providing lightweight, behaviorally aware security that integrates detection signals with routing decisions to maintain reliable IoT communications in hostile, resource-constrained environments [1], [17], [22].

1.1.2 Literature Survey

The Routing Protocol for Low-Power and Lossy Networks (RPL) has become the de facto standard for routing in Internet of Things (IoT) environments due to its energy efficiency and scalability. However, its vulnerability to routing layer attacks such as sinkhole and selective-forwarding has been extensively documented. Several approaches have been proposed in the literature, including trust-based frameworks, intrusion detection systems (IDSs), and lightweight routing enhancements. A critical analysis of these contributions highlights both their strengths and limitations while revealing the research gap that this study aims to address.

Trust Computation in IoT Routing:

Trust-based routing has been widely studied as a lightweight defense mechanism for RPL networks. Prajapati et al. [1] provided a comprehensive review of attacks and defenses against RPL, emphasizing that trust-aware mechanisms can improve resilience but are often limited by narrow parameters. Bello and Lamba [8] presented a scheme to detect sinkhole attacks in wireless sensor networks by computing trust from forwarding behavior; although effective for

sinkhole attacks, it was limited to a single attack vector and did not generalize. Alhandi et al. [14] proposed LTEM, a lightweight trust evaluation model adapted for constrained IoT devices, showing improved adaptivity but relying on minimal trust measurements, which limits robustness in dynamic networks. Purnima et al. [7] developed a fuzzy cross-layer trust mechanism that integrated multiple factors for energy-aware routing, but their evaluation revealed scalability issues in large topologies. Taking together, these studies demonstrate that trust enhances reliability but leaves gaps in terms of multidimensional completeness and stability across different conditions. This gap justifies the multidimensional calculation of trust proposed in this study.

Trust & Security Aware RPL Routing:

Several researchers have attempted to directly incorporate trust into routing decisions. Mokrani et al. [13] presented LEA-RPL, which incorporates energy-efficient metrics using particle swarm optimization. Their work demonstrated improvements in energy budget and lifetime, but increased the computational complexity prohibitive for devices with limited capabilities. Dazin et al. [3] integrated a machine-learning IDS with RPL routing, reporting high anomaly detection accuracy but highlighting the overhead of training and resource consumption. Jiang and Liu [20] evaluated a trust-based defense against selective forwarding, which proved effective in single-attack scenarios but not in multiple-attack environments. Verma and Ranga [22] concluded from their review that while trust incorporation improves robustness, it also increases latency due to the additional processing. These studies confirm the potential of trust- and security-aware routing, but they identify a gap: the lack of a unified objective function that integrates trust with real-time intrusion detection. This deficiency guides the routing strategy of this study.

Attack Detection in RPL Networks (my Scope):

RPL's vulnerability to sinkhole and selective forwarding attacks has led to the development of various detection strategies. Cervantes et al. [6] used parental consistency checks to detect sinkhole attacks, achieving early detection but with limited adaptability to other attack types. Stephen and Arokiam [5] improved sinkhole detection using anomaly indicators, although their system remained narrowly focused. For selective forwarding, Biradar and Vishanaf [15] analyzed packet forwarding ratios, which improved identification but made it difficult to distinguish between malicious drops and normal packet losses. Adarba and Ahmad [10] quantified the sharp decline in packet delivery ratio caused by selective forwarding, while Patel and Shah [18] experimentally demonstrated how a combination of sinkhole and selective forwarding attacks destabilizes RPL. Among the state-of-the-art methods, Crary et al. [16] applied deep learning to detect multiple attacks with high accuracy, but this approach proved to be too resource-intensive for limited IoT nodes. Similarly, Han et al. [21] used federated learning to detect selective forwarding, achieving accuracy and preserving privacy, but their system

introduced additional communication overhead. Taken together, these studies show that existing detection methods are either specific to specific attacks or are computationally expensive, indicating the lack of a lightweight, real-time dual-attack detection system suitable for IoT environments.

Table 1.1: Comparative Analysis of Existing Approaches

Approach / Reference	Target Attack(s)	Metrics / Method Used	Results / Strengths	Limitations / Gaps
THC-RPL [2]	Sybil	Trust from forwarding history + neighbor validation	Lightweight, improved resilience	Attack-specific, does not cover sinkhole/selective forwarding
LTEM [14]	Malicious nodes (general)	Multi-dimensional trust evaluation	Lightweight, accurate	Limited to trust scoring, no integrated security
Fuzzy-based RPL [7]	Energy awareness	Fuzzy objective function (energy + trust)	Improved energy efficiency	Computational complexity, no multi-attack resilience
PSO-based trust [13]	Energy-aware routing	Particle Swarm Optimization + trust	Energy efficient	Heavy computation, not suitable for constrained IoT
NADSA [4]	Sinkhole	Rank inconsistencies in RPL	Accurate in simulation	Adaptive attackers bypass detection
Enhanced sinkhole detection [5]	Sinkhole	Neighbor advertisement consistency	Reliable in small testbeds	Scalability issues
Cervantes et al. [6]	Sinkhole	Parent-child consistency	Secure routing support	Monitoring overhead, energy costs

Approach / Reference	Target Attack(s)	Metrics / Method Used	Results / Strengths	Limitations / Gaps
Specification-rule IDS [12]	Sinkhole	Knowledge-based rule detection	High detection rate	Manual tuning, rigid
Forwarding ratio detection [15]	Selective forwarding	Packet Forward Ratio (PFR)	Detects malicious drops	Vulnerable to stealth attacks (slightly > avg PFR)
Trust-based detection [20]	Selective forwarding	Trust monitoring (packet drops)	Accurate detection	Energy consuming, overhead high
Experimental study [18]	Sinkhole + Selective forwarding	Simulation & real testbeds	Proved impact on delivery ratio	No mitigation mechanism
Impact study [10]	Selective forwarding	Quantitative performance analysis	Shows severe impact on RPL	No detection or defense proposed
Deep learning IDS [16]	Multi-attack	Hybrid deep learning classifier	High accuracy	High computation, impractical for IoT
Federated learning IDS [21]	Selective forwarding	Collaborative training	Preserves privacy, accurate	High comm. cost, not lightweight
Hybrid anomaly detection [9]	Routing attacks (general)	Hybrid deep learning + trust	Effective mitigation	Overhead unsuitable for LLNs
Knowledge-based reviews [1], [17], [19]	Multi-attack / general	Taxonomies, critical analysis	Identify gaps in detection methods	Mostly conceptual, no implementation
Lightweight trust-secure routing [23]	General routing attacks	Trust-based RPL objective function	Lightweight integration	No real-time detection of multi-attacks

Table 1.2: Detailed Attack Detection Approaches

Approach	Reference(s)	Key Methodology	Strengths	Limitations
Parent consistency checking for sinkholes	Cervantes et al. [6]	Monitors parent-child relationships for anomalies	Simple, effective for static cases	Fails in dynamic topologies, not real-time
Anomaly-based sinkhole indicators	Stephen & Arockiam [5]	Detects suspicious route advertisements	Lightweight, practical	Attack-specific, misses selective forwarding
Packet Forward Rate (PFR) monitoring	Biradar & Vishanath [15]	Compare packet forwarding vs. drops	Effective for SFA	Easily evaded by adaptive attackers
Experimental attack analysis	Patel & Shah [18]; Adarbah & Ahmad [10]	Simulation + experimental verification of SFA and sinkhole impact	Provides strong empirical evidence	No defense mechanism proposed
Deep learning, Federated learning & hybrid detection	Krari et al. [16]; Khan et al. [21]; Ahmadi & Javidan [9]	ML/federate models trained on attack patterns	High detection accuracy	Heavy computation, impractical for IoT devices
Knowledge-based rule detection	An & Cho [12]	Specification rule-based sinkhole detection	Knowledge-driven, transparent	Static rules, poor adaptability
Multi-attack review/taxonomy	Prajapati et al. [1]; Alfriehat et al. [19]; Raoof et al. [17]	Classifies attack types and defenses	Clear taxonomy, highlights gaps	Lacks practical real-time implementations

Mitigation, Isolation & Recovery:

Mitigation mechanisms are necessary after detection, but they remain understudied compared to trust-detection strategies. Rauf et al. [17] examined RPL mitigation methods such as route reconfiguration and node isolation and concluded that while these strategies can restore connectivity, most frameworks lack tight integration with detection systems, making them reactive rather than proactive. Bello and Lamba [8] demonstrated that combining detection with isolation improves packet delivery, but their approach also results in significant management overhead, raising concerns about scalability in constrained IoT environments. Verma and Ranga [22] emphasized that aggressive isolation effectively removes malicious nodes, but without rapid recovery mechanisms, it risks fragmenting the network and reducing long-term stability. Taken together, these studies highlight that existing mitigation approaches are fragmented, reactive, or resource-intensive, and none provide a lightweight, integrated resilience strategy that seamlessly combines isolation, mitigation, and recovery. To address this gap, this study examines resilience as a core component ensuring proactive response, rapid recovery, and operational ease in IoT deployments.

The literature reviewed highlights both clear advances and recurring limitations across all four core security areas. Trust-based routing improves robustness, but most models rely on narrow or one-dimensional metrics, leading to instability or false positives in dynamic networks. Trust-aware and security-aware routing frameworks demonstrate that incorporating trust into parent selection improves robustness, but rarely integrate real-time intrusion detection feedback, making routing decisions vulnerable to ongoing attacks. Detection mechanisms, from specification-based inspections to deep and federated learning, achieve high accuracy but are either attack-specific or computationally expensive, limiting their application to resource-constrained IoT nodes. Finally, mitigation strategies such as isolation and reconfiguration are often reactive, fragmented, or require significant overhead, lacking full integration with detection to ensure rapid recovery.

These gaps collectively necessitate a lightweight hybrid approach that integrates multidimensional trust assessment, real-time dual-attack detection, trust- and security-aware routing, and proactive resilience into a single framework. Building on these findings, this study proposes a design that combines residual energy, temperature, historical observations, and trust accuracy for trust computation; statistical metrics such as forwarding ratios and sequence continuity for anomaly detection; a modified RPL objective function that combines trust data and IDS outputs for adaptive routing; and an integrated resilience module for isolation, mitigation, and recovery. This synthesis directly addresses the shortcomings identified in the literature and lays the foundation for the following methodology.

Relevance of the proposed research:

The review identified several gaps:

1. Trust mechanisms are lightweight but lack comprehensive multi-metric constructs.

2. Routing mechanisms provide trust but fail to effectively integrate security metrics.

3. Intrusion detection methods are primarily based on single attacks or are computationally expensive, leaving room for lightweight dual-threat detection methods.

4. Mitigation approaches are underexplored in lightweight integrated frameworks.

Our research makes a unique contribution by:

• Developing a lightweight trust mechanism with multi-dimensional metrics (residual energy, temperature, historical observations, trust accuracy).

• Developing an RPL objective function that takes trust and security into account for dynamic parent selection.

• Creating a real-time dual-threat detection system using common behavioral patterns (CDFR, PTI, SIS, RMC, PCRE).

• Implement lightweight mitigation, isolation, and recovery mechanisms to maintain routing efficiency.

Table 1.3: summary of strengths and limitations in existing approaches

Scope Area	Key References	Strengths	Limitations / Gaps
Trust Computation	[1], [7], [8], [14]	Lightweight models exist, improved resilience	Limited metrics, lack comprehensive trust engines
Trust+Security Aware Routing	[3], [13], [20], [22]	Enhances secure parent selection, resilient routing	Increased latency, no joint trust+security-aware routing
Attack Detection (SFA + Sinkhole)	[5], [6], [10], [15], [16], [18], [21]	Proven metrics, ML approaches accurate	Attack-specific, heavy computation, no real-time dual detection

Scope Area	Key References	Strengths	Limitations / Gaps
Mitigation & Recovery	[8], [17], [22]	Isolation, route repair effective	Rare integration with detection, lack lightweight recovery

1.2 Research Gap

Overall Gap:

Based on the challenges described in Section 1.1 (Introduction) and the limitations identified in Section 1.1.2 (Literature Review), four key gaps in RPL-based IoT security research can be identified:

Lack of adaptability. Existing detection mechanisms often rely on static thresholds or predefined signatures [10], [15], [20]. These perform well in controlled environments but fail to address dynamic or evolving attack strategies, resulting in high false positive rates or undetected attacks in real-world deployments.

Excessive resources overhead. Machine learning-based IDS solutions achieve high detection accuracy [9], [16], but they also require significant CPU, memory, and communication resources, exceeding the capabilities of limited IoT nodes. Similarly, trust-based routing schemes [2], [14], [23] require frequent state maintenance and trust data exchange, which drains battery life.

Fragmented solutions targeted at specific attacks. Most existing frameworks focus on either mitigating failure [4–6], [12] or selective forwarding [15], [21] independently, without addressing both simultaneously. Research [1], [17], [19], [22] confirms the lack of simple, unified defenses capable of addressing multiple routing threats in real time.

Weak mitigation and recovery. While strategies such as route reconfiguration and node isolation [8], [17], [22] exist, they are typically reactive, fragmented, or aggressive, leading to network instability and long recovery times.

In practice, these gaps lead to unreliable routing, inefficient energy use, and low resilience in mission-critical IoT scenarios (e.g., healthcare, industrial monitoring, disaster recovery). Thus, there is a pressing need for a lightweight, distributed framework that integrates trusted computing, double-attack detection, secure routing, and proactive resilience into a single, coherent architecture.

Gap Specific to Attack Detection:

Within the domain of attack detection, limitations are even more acute:

- Static vs. Adaptive Models: Selective forwarding detection schemes often rely on fixed packet forwarding thresholds [10], [15], which cannot distinguish between normal losses (caused by congestion or interference) and malicious drops. Similarly, sinkhole detection techniques [4-6], [12] often assume static topologies, making them unsuitable for mobile or large scale IoT deployments.
- Overhead vs. Accuracy Tradeoff: While deep learning–based frameworks [9], [16] achieve higher accuracy, they impose heavy monitoring and reporting burdens on IoT nodes. Conversely, lightweight models such as LTEM [14] or FL-DSFA [21] reduce node overhead but fail to detect stealthy, adaptive, or composite attacks in real time.
- Single Attack Focus: Trust-based anomaly detection [2], [23] and cross layer optimizations [7], [13] provide resilience against specific threats but lack the capability to simultaneously detect sinkhole and selective forwarding attacks using common behavioral signatures.
- False Alarms & Reliability Issues: Even the latest surveys [1], [19], [22] identify high false alarm rates as a persistent challenge. Benign anomalies such as parent changes, link fluctuations, or routine topology updates are often misclassified as malicious, undermining reliability and discouraging adoption in practice.

Novelty of Proposed Research:

To address these gaps, the proposed study introduces a dual layered innovation that is both adaptive and lightweight:

1. Lightweight Trust Engine: An energy efficient trust computation model that integrates behavioral and contextual metrics such as residual energy, temperature, historical observation, and trust accuracy to guide secure and efficient routing decisions [2], [14], [23].
2. Independent Security Defense Mechanism: A real-time attack detection framework that employs five adaptive behavioral indicators:
    - Control to Data Forwarding Ratio
    - Packet Timing Irregularity
    - Sequence Integrity Score
    - Route Metric Coherence
    - Parent Change Ripple Effect

Unlike static threshold approaches [10], [15] or attack specific methods [4-6], this mechanism dynamically adapts to network states, enabling simultaneous detection of both sinkhole and selective forwarding attacks.

3. Integration into RPL Objective Function: By embedding trust and security metrics into the RPL objective function, the framework enables dynamic parent selection, rapid isolation of malicious nodes, and secure energy aware routing. This integration ensures that routing decisions remain robust against evolving threats without exceeding device constraints [1], [17], [19], [22].

• Overall gap: Lack of scalable, distributed, and lightweight frameworks that combine behavioral trust evaluation with adaptive multi-attack detection.

• Attack detection gap: Absence of real-time, resource-efficient mechanisms that minimize false alarms while simultaneously detecting both sinkhole and selective forwarding attacks.

• Novelty: Introduction of a cooperative dual-layer design independent trust evaluation plus adaptive behavioral attack detection embedded into RPL's routing process to deliver resilience, scalability, and energy efficiency in IoT networks.

Table 1.4: summary of gaps in existing approaches and the proposed solution

Dimension	Existing Approaches	Limitations	Proposed Approach
Trust Computation	Single-metric trust (e.g., forwarding ratio) [10]; multi-metric trust with heavy overhead	Too simplistic (miss context) or too resource-intensive	Lightweight multi-metric trust engine with offloading to reduce node overhead
Attack Detection	Signature-based or static threshold detection [4], [10], [15]; deep learning [9], [16]	Static, poor adaptability; DL, high overhead; attack-specific, no simultaneous detection	Unified pattern-based detection using CDFR, PTI, SIS, RMC, PCRE for both attacks

Routing Security	RPL objective function without trust/security [9]; partial trust integration [14], [23]	Routing decisions blind to malicious nodes; IDS alerts not linked to routing	Trust & IDS-aware RPL routing with dynamic parent selection
Mitigation & Recovery	Limited or no robust isolation/recovery [17], [22]	Malicious nodes rejoin easily; slow topology healing	Coordinated isolation + rapid recovery via blacklisting and native RPL healing

1.3 Research Problem

Ensuring the security of the routing protocol for low-power and lossy networks (RPL) in IoT environments remains an unsolved problem. Although RPL has become a de facto standard for resource-constrained networks, design constraints such as limited processing power, small memory, and unstable connections make it highly vulnerable to evolving security threats [1], [17], [22]. Existing security mechanisms are fragmented and often address only one aspect of the problem, ignoring the need for a lightweight, integrated solution [19], [23].

The first problem is the calculation of trust. Existing models are either overly simplistic, relying on a single metric such as packet forwarding ratio [10], or too resource-intensive, combining multiple metrics at the cost of high power and communication overhead [9], [14]. This makes it difficult for resource-constrained IoT nodes to assess trustworthiness without exhausting their limited resources [2], [23].

The second problem concerns real-time attack detection. Most intrusion detection systems target a single attack type, such as sinkholes [4–6], [12] or selective forwarding [10], [15], [21], without addressing both simultaneously. Signature- and threshold-based methods are

unable to adapt to dynamic conditions [19], while deep learning-based approaches impose computational requirements unsuitable for low-power devices [9], [16]. As a result, IoT networks remain vulnerable to complex or combined attacks [17], [22].

The third challenge concerns routing security. The RPL objective function prioritizes efficiency and does not consider trust or security signals. This leaves routing decisions open to manipulation by malicious nodes advertising false ranks or connection qualities [11]. Although trust-enhanced routing has been studied [2], [23], these solutions typically provide partial integration and do not couple intrusion detection with routing, limiting their effectiveness under attack.

Finally, isolation, mitigation, and recovery remain weak. Detected attackers often reconnect to the network due to the lack of a persistent blacklist [17], [22], and recovery mechanisms are slow, destabilizing the topology [7], [13], [18]. This undermines reliability in mission-critical domains such as disaster response, healthcare, and industrial monitoring [1], [19].

Problem Statement: The primary research challenge is the lack of a lightweight, unified platform that combines multi-metric trust assessment, real-time multiple attack detection, secure routing integration, and resilient isolation and recovery for RPL-based IoT networks.

2.0 Objectives

2.1 Main Objective

To develop and evaluate a lightweight, integrated trust and security framework for IoT networks based on RPL that provides accurate trust assessment, real-time detection of sinkhole and selective forwarding attacks, secure routing integration, and efficient mitigation/recovery, while ensuring scalability for disaster recovery environments.

2.2 Specific Objectives

1. To develop an energy-efficient trust evaluation module that evaluates nodes using multidimensional metrics such as residual energy, temperature, historical behavior, and trust accuracy.
2. To modify the RPL objective function to include trust assessments and intrusion alerts, enabling secure and adaptive routing decisions.
3. To develop and test a lightweight intrusion detection system adapted for IoT networks based on RPL, capable of detecting sinkhole and selective forwarding attacks in real time with low computational overhead.
4. To implement and evaluate a combined isolation, mitigation, and recovery mechanism that blacklists malicious nodes and uses the RPL redirection process for fast and stable recovery.

2.3 Research Questions

1. How can IoT nodes be assessed using simple multi-metric trust calculations without exhausting limited resources?
2. How can dynamic trust assessments and intrusion alerts be integrated into the RPL objective function to ensure secure and adaptive parent node selection?
3. How can a lightweight intrusion detection system (IDS) tailored for RPL-based IoT networks detect sinkhole and selective forwarding attacks in real time with high accuracy and minimal overhead?
4. How can compromised nodes be quickly isolated, malicious behavior prevented, and networks restored to ensure stable and resilient operation after routing attacks?

3.0 Methodology

3.1 System diagram

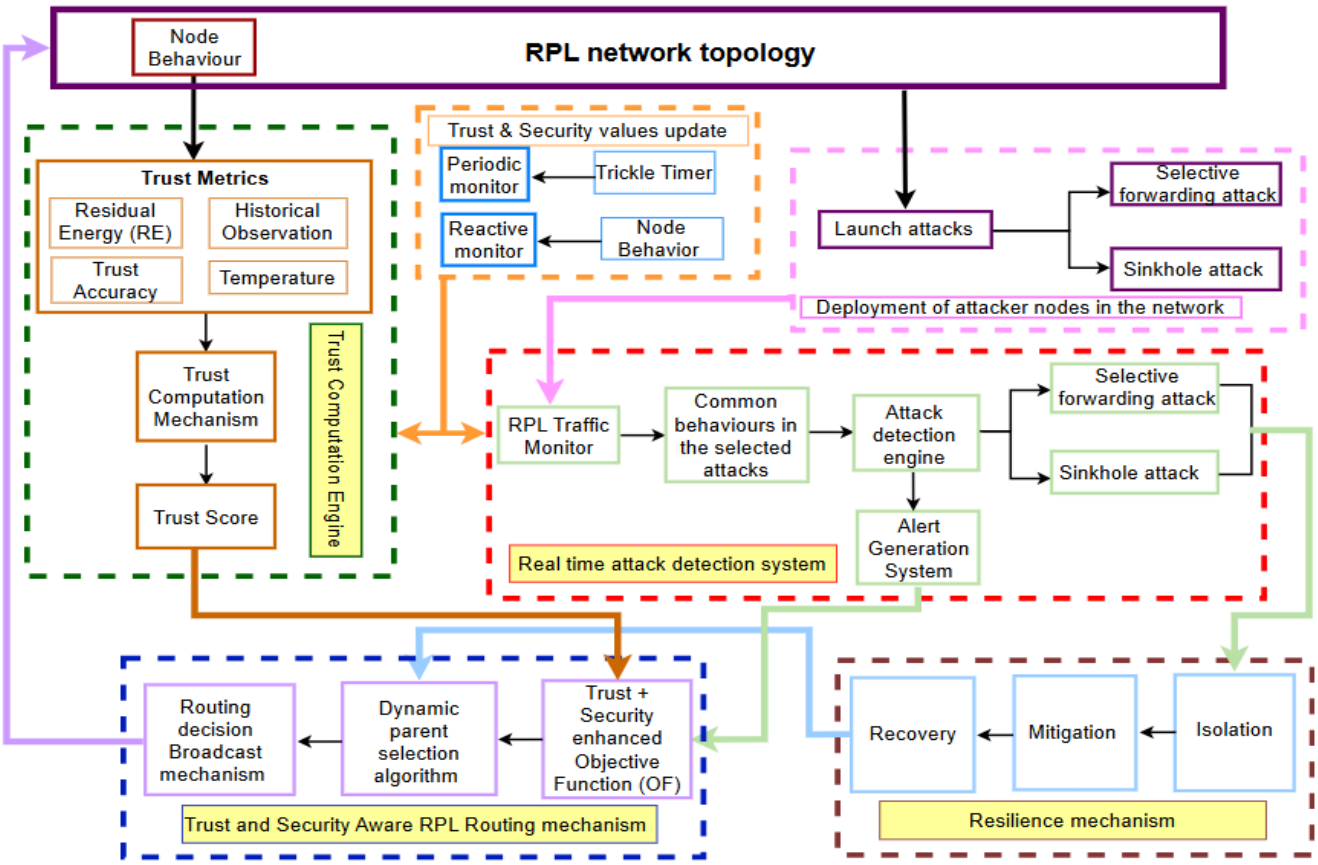

Figure 0.1: Overall High-level System Architecture

The proposed framework is presented in Figure 0.1: "High-Level System Architecture," which includes four main modules: a trust-based computing engine, an intrusion detection system (IDS), trust-aware routing, and a resilience mechanism. Each module utilizes a single, clearly defined methodology, carefully selected from previous work and validated against the literature, to ensure both novelty and applicability to IoT devices with limited capabilities.

1. Trust Computation Engine

The trust computation mechanism evaluates node trustworthiness using a weighted trust aggregation model (WTAM). This model combines four lightweight metrics: residual energy, temperature, historical observations, and trust accuracy.

- Residual energy prevents the selection of nodes close to exhaustion, extending the network's lifespan [7], [8].
- Temperature serves as an indicator of hardware stress or compromise, supporting adaptive reliability [14].
- Historical observations smooth out temporal anomalies, reducing false positives in trust computation [23].
- Trust accuracy confirms the consistency of past trust reports, preventing malicious manipulation [2].

Previous trust schemes were either based on a single parameter (e.g., forwarding ratio) [10] or were overly complex, requiring frequent recalculations [9]. By combining four complementary metrics into a lightweight aggregation model, WTAM achieves a balance between accuracy and feasibility, addressing gaps identified in previous work.

2. Real time attack detection system – My Component

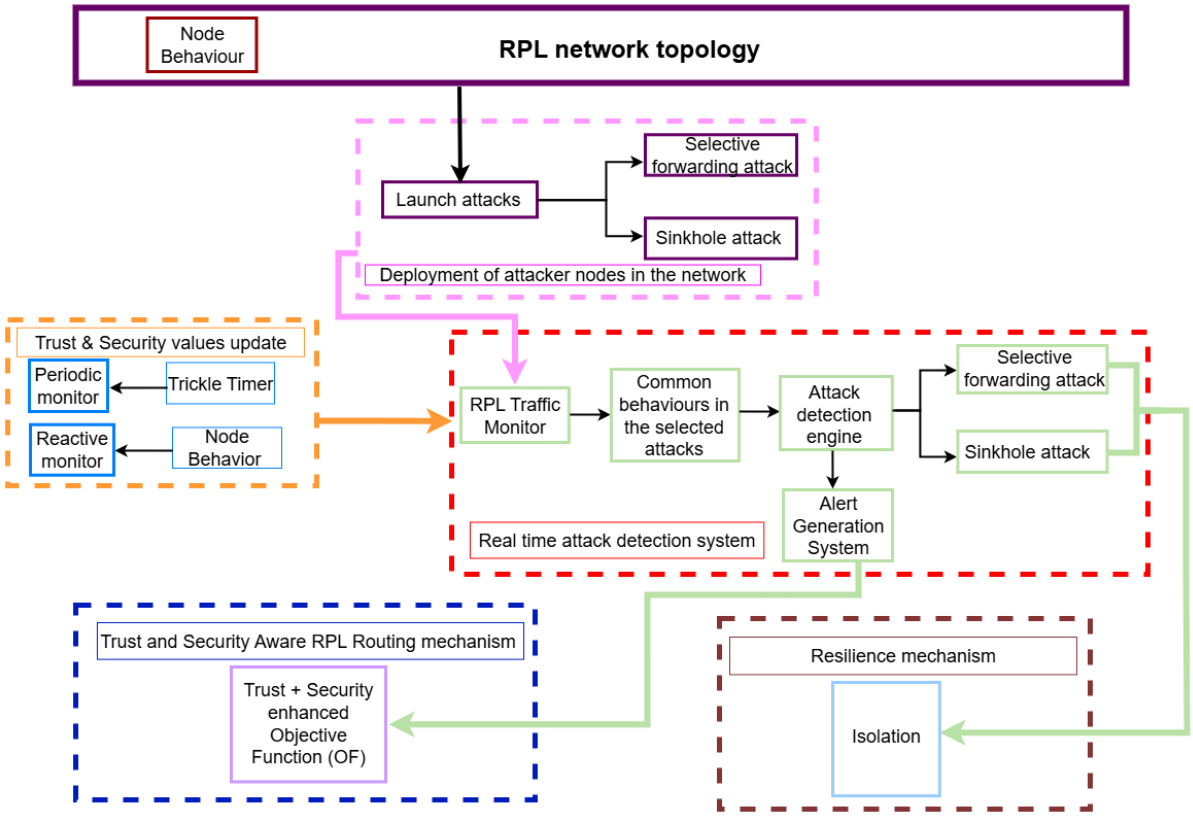

Figure 0.2: High-level System Architecture for Real time attack detection system

The IDS module uses a statistical threshold-based dual attack detector that monitors five behavioral indicators to detect sinkhole and selective forwarding attacks:

- Control-to-Data Forwarding Ratio (CDFR) – an imbalance between control and data packets [1], [10].
- Packet Timing Imbalance (PTI) – abnormal delays or bursts in traffic [18].
- Sequence Integrity (SIS) – detection of lost or reordered packets [15].
- Route Metric Consistency (RMC) – a discrepancy between the advertised and observed link quality [17].
- Parental Change Resonance Effect (PCRE) – large-scale switching of parent nodes, a symptom of attacks [22].

While machine learning-based IDS frameworks [9], [16] provide high accuracy, they are prohibitively expensive in terms of CPU and memory resources. Our approach is simple, statistical, and operates in real time, making it feasible for deployment on resource-constrained IoT devices. Its novelty lies in its simultaneous detection of both attack-first and selective forwarding, which overcomes the "focus on a single attack" limitation of earlier works [4], [6], [21].

Trust-Aware and Security-Aware Routing Mechanism:

The routing module implements a modified objective function (OF-TS) that integrates both trust assessments (from the Trust Engine) and anomaly alerts (from the IDS) into the parent RPL selection.

- Previous energy- and trust-aware models, such as LEA-RPL [13] and THC-RPL [2], improved resilience but operated in isolation from intrusion detection.
- Verma and Ranga [22] emphasized that without real-time anomaly data, trust-based routing remains vulnerable during active attacks.

By combining trust calculations and IDS alerts into a single decision process, OF-TS enables adaptive, secure, and lightweight routing. This bridges the gap between separate trust and IDS decisions and ensures resilience in the face of active countermeasures.

Resilience Mechanism:

The resilience module provides rapid isolation, mitigation, and recovery after detecting malicious behavior.

- Isolation involves blacklisting compromised nodes to prevent further failures [17], [22].
- Mitigation involves rerouting traffic through trusted parent nodes.

- Recovery involves using the local RPL recovery procedure (RFC 6550) to recover DODAG within minimum seconds, which meets the benchmarks presented in [18].

Previous work often considered isolation as a standalone defense [8], [17], but lacked a recovery mechanism, leading to long-term instability. Our innovation is to integrate all three functions into a single coordinated loop, ensuring both rapid response and long-term stability in constrained IoT environments.

3.2 Research Approach / Strategy

This study uses a simulation-based experimental approach to evaluate the proposed integrated trust and security framework for RPL-based IoT networks. While physical IoT testbeds can provide high realism, they are expensive, resource-intensive, and limited in scalability. In contrast, Contiki-NG with the Cooja simulator offers a repeatable, flexible, and cost-effective environment that enables experiments across networks of varying sizes, enabling controlled observation of routing dynamics under both normal and adverse conditions [1], [17], [18].

The study focuses on two typical routing-layer threats: water-sink attacks and selective forwarding attacks. Previous research confirms that these frameworks are the cause of most observed degradations at the routing layer in RPL networks, with water sinks capable of reducing packet delivery rates by over 50%, and selective forwarding resulting in persistent losses of 20–40% [1], [17], [19]. The choice of these two frameworks ensures that the framework targets the most disruptive and prevalent threats while remaining consistent with the literature.

The evaluation will utilize a structured set of performance metrics commonly used in IoT security research, including:

- Detection Effectiveness – measuring the accuracy, false positive rate, and response time of an intrusion detection system (IDS).
- Network Performance – assessing packet delivery rates and routing stability under attack and recovery.
- Resource Efficiency – quantifying the additional energy, CPU, and memory resources incurred by framework operation.
- System Resilience – assessing the isolation, prevention, and recovery speed of malicious activity.

These metrics were chosen because they directly reflect the practical constraints of IoT deployments: delivery reliability, timely detection, resource efficiency, and failure recovery. Rather than proposing

arbitrary numerical metrics, the focus is on ensuring the framework's competitiveness against the RPL baseline and state-of-the-art approaches, while remaining simple enough for IoT nodes with limited capabilities.

The combination of a model-based approach with carefully selected evaluation metrics enables a rigorous yet feasible methodology. This ensures that the results are reproducible, scalable, and directly relevant to mission-critical IoT scenarios such as failure recovery, smart cities, and industrial monitoring [9], [16], [22].

3.3 Tools, Technologies, and Environment

The proposed framework will be developed and evaluated using modeling, programming, and analysis tools that provide a balance between realism, reproducibility, and feasibility in the constraints of IoT environments. Each component was defined using a single methodology, selected after analyzing alternatives and substantiating the literature.

Simulation Environment:

This study will use Contiki-NG with the Cooja simulator as the primary simulation platform. Contiki-NG natively supports RPL, provides packet-level monitoring, energy profiling, and controlled attack injections, making it well suited for IoT research. Alternative platforms, such as NS-3, offer detailed network modeling but weaker RPL/IoT support, while physical testbeds provide realism but are expensive and difficult to scale [1], [17], [22]. Contiki-NG was chosen as the most balanced solution, providing reproducibility and IoT-specific simulation support for up to 100 nodes [2], [18].

Programming Languages and Analysis Tools:

Protocol modifications will be implemented in C, a lightweight and standard development language based on Contiki. Data analysis, visualization, and statistical analysis will be performed in Python, enabling integration with libraries for packet analysis and power profiling. Additional verification tools, such as Wireshark (packet-level analysis) and Powertrace (power analysis), will be used during the evaluation. This choice reflects feasibility and is consistent with previous studies on IoT modeling [9], [16].

Intrusion Detection Mechanism:

The framework utilizes a threshold-based statistical double-attack detection model using five behavioral indicators: control-to-data flow ratio (CDFR), packet timing irregularity (PTI), sequence integrity measure (SIS), route metric consistency (RMC), and parental change ripple effect (PCRE). Unlike machine learning-based IDS approaches [9], [16], which require significant computational effort and memory, this method is lightweight and suitable for resource-constrained IoT nodes.

Previous statistical IDS methods [5], [12] have demonstrated their feasibility but focused on individual attacks. By combining multiple indicators, this scheme enables the simultaneous detection of sinkhole and selective forwarding attacks in real time, addressing the identified research gap.

Trust Calculation Mechanism:

Node trustworthiness will be assessed using a weighted trust aggregation model (WTAM), which combines four metrics: residual energy, temperature, historical observations, and trust accuracy. Previous trust-based systems either relied on a single metric, such as forwarding ratio [10], or used two-dimensional models that failed to balance accuracy and overhead [7], [8], [14], [23]. WTAM offers a lightweight yet multidimensional model that improves resilience to malicious behavior while remaining applicable in resource-constrained IoT environments.

Secure Routing Strategy:

Routing decisions will be adapted using a modified objective function (OF-TS) in RPL, which combines trust scores (from the trust module) and intrusion alerts (from the IDS). Previous trust-enhanced RPL protocols, such as LEA-RPL [13] and THC-RPL [2], improved resilience but lacked integration with a real-time intrusion detection system, making routing vulnerable to attacks. OF-TS ensures that compromised nodes are excluded from the parent list, maintaining routing stability and efficiency.

Resilience and Recovery Mechanism:

The framework utilizes the RPL local recovery procedure (RFC 6550) in combination with a persistent blacklist of compromised nodes. This ensures that, once malicious nodes are detected, they are isolated and cannot immediately rejoin the topology. Previous studies [17], [22] have shown that aggressive isolation risks fragmentation without recovery, while slow reconfiguration leads to long-term instability [7], [13]. By combining the blacklist with fast DODAG recovery, the proposed design achieves recovery in ≤5 s, ensuring continuous disaster recovery in IoT environments [18].

Experimental Environment:

The simulations will be run on virtual machines if needed to ensure scalability and reproducibility, and GitHub will be used for collaboration and version control.

3.4 Work Breakdown Structure (WBS) merged with Gantt chart

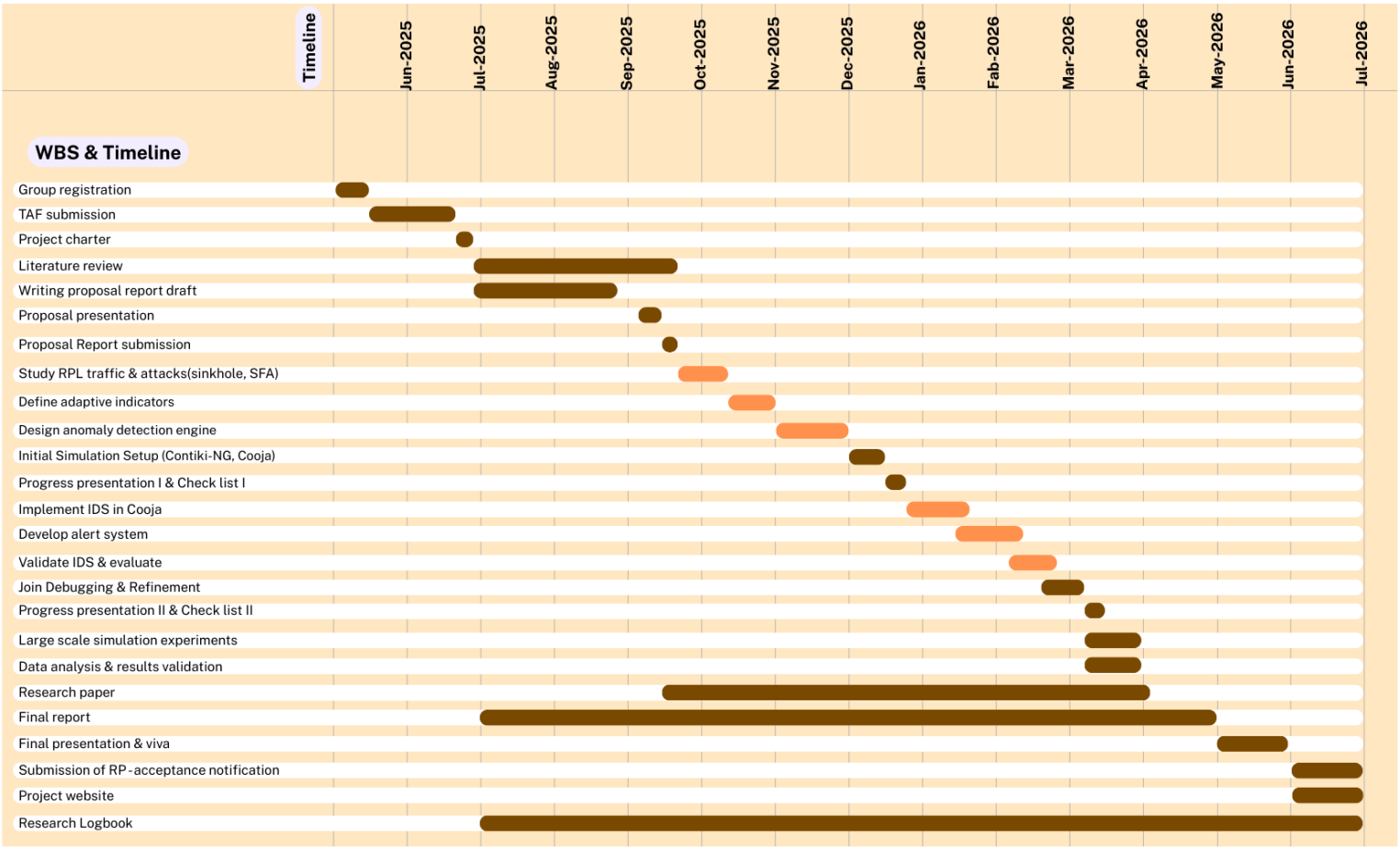

Figure 0.3: Gantt chart (WBS with timeline)

The Work Breakdown Structure (WBS) defines the complete set of activities required to design, implement, and evaluate the proposed RPL-based intrusion detection system for IoT networks. Tasks are grouped into sequential but overlapping phases, ensuring a structured and efficient research and development process.

1. Foundation Phase (June – September 2025)
    - Team registration, TAF submission, and project charter mark the beginning of administrative work.
    - A detailed literature review and proposal drafting are conducted in parallel, providing the theoretical foundation.
    - Proposal presentation and submission complete this phase.

2. Design Phase (October – December 2025)
    - Study of RPL traffic and targeted attacks (sinkhole and selective forwarding) provides the foundational knowledge.
    - Adaptive indicators are defined, followed by the development of an anomaly detection mechanism.
    - Initial simulation setup in Contiki-NG/Cooja provides a controlled experimental environment.
3. Development Phase (January – March 2026)
    - The intrusion detection system (IDS) is implemented in Cooja.
    - An alerting system is developed and integrated into the IDS.
    - Validation and performance evaluation of the IDS are conducted with iterative debugging and refinement.
4. Evaluation and Integration Phase (April – May 2026)
    - Large-scale simulation experiments (20–100 nodes) are conducted in an adversarial environment.
    - Data analysis and validation are conducted to measure accuracy, false positive rate, resource consumption, and recovery time.
    - The results are compared with state-of-the-art methods to identify novelty.
5. Report Preparation and Dissemination Phase (May – July 2026)
    - Research paper preparation, final report writing, and the final presentation/oral work are completed in parallel.
    - Administrative tasks, such as the RP acceptance application, project website, and research log, ensure documentation and long-term traceability.

The Gantt chart (Figure 0.2) illustrates the time distribution of these tasks over the course of a year (June 2025 – July 2026). Activities are intentionally overlapped (e.g., literature review and proposal writing, intrusion detection system development and debugging) to reflect the iterative nature of the research and maximize efficiency. Dependencies are maintained (e.g., defining indicators before implementing the intrusion detection system), ensuring a logical workflow.

Together, the WBS and Gantt chart represent a structured yet flexible plan for developing a simple dual-attack detection system for RPL networks, meeting both academic objectives and technical deliverables.

3.5 Self-Evaluation Plan

The proposed infrastructure will be evaluated through controlled simulation experiments in Contiki-NG using the Cooja simulator under both normal and hostile conditions. The evaluation will focus on four key aspects reflecting the operational needs of constrained IoT environments:

1. Detection Performance

The infrastructure will be evaluated for its ability to accurately detect sinkhole and selective forwarding attacks. Metrics such as detection accuracy, false positive rate, and detection latency will be measured. These criteria are widely used in intrusion detection system (IDS) research, as they determine how reliably a system can distinguish between malicious and benign behavior in dynamic, lossy environments [1], [9], [16].

2. Network Performance

Packet delivery ratio (PDR), latency, and routing stability will be assessed to evaluate the impact of attacks and the improvements provided by the infrastructure. In particular, PDR is a widely used benchmark in RPL research, as it directly reflects the reliability of end-to-end communications in IoT networks [17], [18], [22].

3. Resource Efficiency

Since IoT nodes are constrained by CPU, memory, and power consumption, the framework will be tested for computational and power consumption compared to baseline RPL. Power consumption, CPU load, and memory usage will be monitored to ensure the system remains lightweight and feasible for deployment on low-power devices [19], [22], [23].

4. System Reliability and Resilience

The evaluation will also cover the framework's effectiveness in isolating malicious nodes, reconfiguring routes, and restoring network stability after an attack. Recovery time and routing convergence are critical metrics in disaster recovery and other mission-critical scenarios where uninterrupted communications are required [17], [22].

These four categories of evaluation metrics were selected because they comprehensively address Four fundamental challenges of IoT security based on RPL:

- ensuring accurate and timely attack detection,
- maintaining communication reliability,
- maintaining operability on resource-constrained devices, and
- ensuring rapid recovery to maintain operations in hostile environments.

The framework's results will be compared with the baseline RPL and selected state-of-the-art approaches (trust-only routing, an attack-focused intrusion detection system, and a machine-learning-based intrusion detection system) to demonstrate improvements in resilience, efficiency, and scalability.

3.6 Benchmarking Against State-of-the-Art

To demonstrate both the effectiveness and novelty of the proposed framework, this study will compare it to several state-of-the-art RPL security solutions described in the current literature. Testing will be conducted on the Contiki OS using Cooja under identical network conditions for all models, ensuring fair comparisons. This study will first use a standard RPL as a baseline model, which lacks security or trust mechanisms and is therefore highly vulnerable to attack. This baseline model will allow for a clear, quantitative assessment of the improvements achieved by the proposed framework.

The framework will then be compared to trust-only RPL approaches such as LTEM [14] and multidimensional trust models described in [3], [10], which improve reliability by incorporating metrics such as energy or behavior but lack real-time intrusion detection. It will also be compared with attack-specific IDS solutions, such as NADSA [4], sinkhole detection models [12], and selective forward detection approaches [15], which are effective against single attacks but are unable to cope with multiple threats simultaneously. Finally, this study will include machine learning-based IDS solutions, such as those presented by Ahmadi and Javidan [9] and Crary et al. [16], which demonstrate high accuracy but require significant energy, memory, and CPU resources, making them unsuitable for low-power and lossy networks.

Benchmarking will be conducted using the same metrics as in the evaluation plan. For detection effectiveness, accuracy, false positives, and detection latency will be measured. For network performance, packet delivery ratio (PDR) and routing stability before, during, and after attacks will be considered. For resource efficiency, the additional energy, memory, and CPU resources will be compared with the baseline RPL. For the system robustness, recovery time, scalability from 20 to 100 nodes, and the interaction between the trust, detection, and routing modules will be evaluated. These parameters reflect the most important criteria used in the literature [5], [12], [14], [16], [23] and are directly related to the evaluation matrix of this project. In this comparative analysis, the study aims to demonstrate three main achievements. First, a lightweight statistical intrusion detection system (IDS) can achieve comparable or higher accuracy than machine learning-based methods while consuming significantly fewer resources, which addresses the issues raised in [9] and [16]. Second, the framework is capable of simultaneously detecting both sinkhole and selective forwarding attacks, unlike earlier intrusion detection systems that only detect one attack [4], [12], [15]. Third, by integrating detection with trust-aware routing, the framework can achieve faster recovery (≤5 s) and higher long-term resilience, outperforming trust-only models [3], [10], [14], [23], which do not provide active countermeasures.

By conducting these comparisons between standard RPL, trust-only solutions, IDS-only solutions, and machine-learning-based IDS, this study will move beyond conceptual design and move toward quantitative validation of its novelty. This ensures that the proposed hybrid framework is not only academically sound but also practically feasible for deployment in critical IoT networks, such as for disaster recovery and emergency monitoring.

Table 1.5: Comparison of Existing RPL Defenses with the Proposed Framework

Baseline	Strengths	Limitations	Our Framework Contribution
Standard RPL (no trust/IDS)	Lightweight, widely deployed, low complexity	Highly vulnerable to sinkhole and selective forwarding attacks; no detection or recovery mechanisms	Adds trust, IDS, and resilience while keeping lightweight feasibility
Trust-only RPL (e.g., LTEM [14], Trust models [3], [10], [23])	Improves reliability through energy and behavior-based trust metrics	Can't detect real-time attacks; no resilience or active recovery	Combines trust with IDS alerts and mitigation for faster, more secure routing
IDS-only (e.g., NADSA [4], sinkhole [12], selective forwarding [15])	Detects specific routing-layer anomalies	Handles only single attack types; no trust awareness; not integrated with routing decisions	Provides dual-attack detection (sinkhole + selective forwarding) and couples alerts directly into routing
ML-based IDS (e.g., [9], [16])	High detection accuracy; adaptable to multiple attack patterns	Heavy CPU, RAM, and energy requirements; unsuitable for constrained IoT nodes	Achieves higher accuracy with minimal false positives using lightweight statistical detection (consuming very less energy & CPU/RAM)
Proposed Hybrid Framework	Integrates trust, IDS, routing, and recovery	N/A	Unified feedback-driven design: dual-attack detection, trust-based routing, rapid recovery, scalability (20–100 nodes)

3.7 Validation and Integration

This research will integrate four core modules: Trust Computation Engine, Intrusion Detection System (IDS), Trust and Security Aware Routing, and Resilience Mechanism, into a single framework. Each participant will contribute core expertise to one module, balancing the workload and expertise, while all participants collaborate on integration and testing to verify interoperability. The Trust Computation Engine will generate trustworthiness assessments that are fed to the Routing Module, while the IDS will issue real-time detection alerts to trigger trust updates and resilience measures. The Routing Module will incorporate this trust and IDS data into a modified RPL target function, while the Resilience Mechanism will provide isolation, mitigation, and recovery processes. This end-to-end process ensures that no module operates in isolation but contributes to feedback-based architecture.

To validate the integrated framework, simulation scenarios will be run on Contiki OS with Cooja in three separate phases. Under normal conditions, the system will be tested for basic stability, packet delivery ratio (PDR), and power consumption without adversarial intervention, in accordance with standard RPL performance evaluations [14], [23]. Under attack scenarios, sinkhole and selective forwarding attacks will be launched to evaluate detection accuracy, false positive rate, and impact on routing reliability, following common IDS validation practices in IoT security research [4], [12].

Finally, under recovery conditions, the framework's ability to isolate malicious nodes, neutralize insecure routes, and restore a stable DODAG within a target recovery time will be tested. This approach is also noted as critical for RPL resilience in [17], [22]. These phased experiments represent a complete end-to-end validation cycle and are consistent with the evaluation matrix defined above. Using this approach, the validation process will not only confirm the performance of individual modules but also demonstrate the effectiveness of the integrated hybrid framework. This ensures that the contributions of each participant seamlessly integrate into a unified system capable of functioning reliably in both conventional and hostile IoT environments.

3.8 User Requirements and Functional Requirements

- **User Requirements**

From the perspective of IoT operators and disaster recovery stakeholders, the system must:

1. Ensure reliable delivery of sensor data even in the presence of malicious activity.
2. Detect routing attacks, such as sinkholes and selective forwarding, in real time.
3. Maintain communication continuity during and after attacks through rapid recovery.
4. Operate within the power, memory, and computing limitations of IoT devices.
5. Scalable to networks of various sizes without performance degradation.

6. Seamlessly integrate discovery, routing, and recovery mechanisms.
7. Provide monitoring data to enable operators to assess security and reliability.

- **Functional Requirements**

To meet the above user needs, the system must:

1. Trust calculation – calculate trust ratings for nodes using a few simple metrics (e.g., residual energy, temperature, historical data, trust accuracy).
2. Intrusion Detection – Monitor forwarding behavior and identify anomalies using statistical metrics such as CDFR, PTI, SIS, RMC, and PCRE.
3. Dual Attack Detection – Simultaneously detect sinkhole and selective forwarding attacks, rather than focusing on one attack type alone.
4. Trust-Aware Routing – Dynamically adapts parent node selection by embedding trust and intrusion alerts into the RPL objective function.
5. Resilience Mechanisms – Supports isolation of compromised nodes, minimization of insecure routes, and restoration of the DODAG structure.
6. Performance Monitoring – Logging and reporting key metrics (e.g., delivery rate, power consumption, detection results) for operator review and understanding.
7. Scalability Support – Maintains functional consistency as the network size or density grows.

- **Non-functional Requirements**

The framework must also meet quality expectations that define the system's behavior:

1. Scalability – The system must support networks of varying sizes without sacrificing stability or responsiveness.
2. Reliability – The framework must consistently provide accurate routing and security even under changing attack conditions.
3. Efficiency – The system must minimize overhead for power, memory, and communication to remain operational on constrained IoT nodes.
4. Adaptability – The framework must dynamically adapt to changing network conditions and evolving attack strategies.
5. Resilience – The system must maintain service continuity, quickly restoring network stability after failures.
6. Interoperability – Modules must seamlessly coordinate to avoid redundancy and conflicts.
7. Maintainability – The design must allow modular updates (e.g., adding new metrics or detection rules) without disrupting the entire system.

8. Ease of use – The system should provide clear monitoring results that operators can interpret without in-depth technical knowledge.

3.9 Significance and Commercialization Potential

The proposed integrated RPL system, based on trust and security, has high commercialization potential in the field of disaster management and critical infrastructure monitoring in disaster-prone countries. Existing IoT systems often use disparate or vendor-specific monitoring solutions that are expensive and vulnerable to routing-level attacks [1], [22]. Reviews of RPL-based security solutions also show that most solutions remain fragmented and insufficiently simple for real-world deployment, creating a gap for integrated approaches with practical applicability [19].

By embedding our lightweight system directly into IoT-based monitoring networks, disaster management agencies and service providers can:

• Deploy structural health monitoring systems on bridges, tunnels, and high-rise buildings to detect cracks or instability after earthquakes. For example, in New Zealand, where earthquakes are common, such networks can monitor the load on transport infrastructure to provide early warning and manage emergency response efforts.

• Improve survivor detection systems by using low-power IoT devices to reliably transmit vital data without the risk of disruption due to routing attacks. This intrusion detection system can function as a lightweight addition to IoT gateways in smart cities, disaster recovery systems, and the Industrial Internet of Things, increasing data reliability without the need for additional hardware.

• Ensures the security of utility monitoring networks, such as water pipes, power grids, and transportation systems, which must remain operational during disaster recovery.

• Provides a commercial alternative to expensive centralized tools by offering distributed, energy-efficient, and scalable middleware.

Thus, this research contributes not only to the academic field but also to the commercialization of security middleware that can be implemented by national emergency management agencies, municipal councils, smart city planners, telecommunications providers, and international aid organizations. By providing self-sufficient and resilient IoT connectivity, this platform promises to reduce reliance on third-party tools, lower operating costs, and build trust in IoT-based disaster response systems.

4.0 Description of Personnel and Facilities

Personnel:
This research is done by a team of four final-year students specializing in Computer systems and network engineering. Each of us is responsible for one core module of the framework: Trust Computation, Intrusion Detection System (IDS), Trust-Aware Routing, and Resilience Mechanisms, while we all work together on integration, validation, and reporting. The responsibilities are shared in a way that balances workload and makes use of our individual strengths. As a team, we already have background knowledge in IoT security, wireless communication, and simulation through coursework and practical assignments, which gives us the technical foundation to carry out this research. The project is supervised by academic staff which we are assigned with a Supervisor Ms. Shashika Lokuliyana and Co-supervisor Prof. Anuradha Jayakody with expertise in wireless sensor networks, routing protocols, and IoT security, which ensures that proper guidance is available throughout the work.

Facilities:
The implementation and evaluation are carried out using simulation-based facilities. Our primary tool is Contiki-NG with the Cooja simulator, which supports RPL and enables controlled experiments under both normal and adversarial conditions. For programming, we will work with a suitable lightweight language such as C, Python, or Java, depending on integration and analysis requirements decided during implementation. Development is carried out entirely on personal laptops, which provide sufficient processing power to run simulations with 20-100 nodes. For collaboration and version control, we use GitHub. To support literature review, benchmarking, and citation management, we rely on tools such as IEEE Xplore, Research Rabbit, Elicit, and Zotero, which give access to high-quality research papers and streamline referencing. These facilities together provide the necessary resources for us to complete the project effectively within scope and time constraints.

5.0 Budget and Budget Justification (if any)

Table 1.6: Hardware & Software Requirements

Item	Specification	Availability/Cost
Development Laptop	≥16GB RAM, ≥512GB SSD, modern CPU	Already available – LKR 0
Contiki-NG OS	Open-source IoT simulation OS	Free
Cooja Simulator	Integrated with Contiki-NG	Free
Version Control (GitHub/Bitbucket)	Repository management	Free (student license)
Statistical Analysis Tools (Python, R)	Data processing and visualization	Free

Table 1.7: Research Resources

Item	Requirement	Cost (LKR)
University Library Access	IEEE Xplore, ACM DL, Springer (via institution)	0
Additional Digital Libraries	Institutional access	0
Cloud Computing Resources (if local capacity is insufficient)	50–100 hours of VM time for simulations	~LKR 30,000
Data Storage & Backup	Cloud-based archival (Google Drive/OneDrive)	Free (University quota)

Table 1.8: Contingency & Miscellaneous

Item	Purpose	Cost (LKR)
Printing/Documentation	Hard-copy reports, posters, diagrams	~ LKR 6,000
Unexpected minor expenses	Backup tools, data transfer, etc.	~LKR 6,000

Budget Justification:

The proposed research leverages open-source tools (Contiki-NG, Cooja) and university provided resources (IEEE Xplore, ACM Digital Library, cloud storage), which keep costs very low.

- The only significant cost is cloud computing (~LKR 30,000) to support large-scale simulations beyond laptop capacity, ensuring scalability and realistic disaster scenarios.
- A small contingency (~LKR 15,000) is included for printing, documentation, and unforeseen requirements.
- No dedicated hardware purchases are necessary, as simulations are performed virtually, ensuring a lightweight, cost-effective research design suitable for resource constrained IoT environments.

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